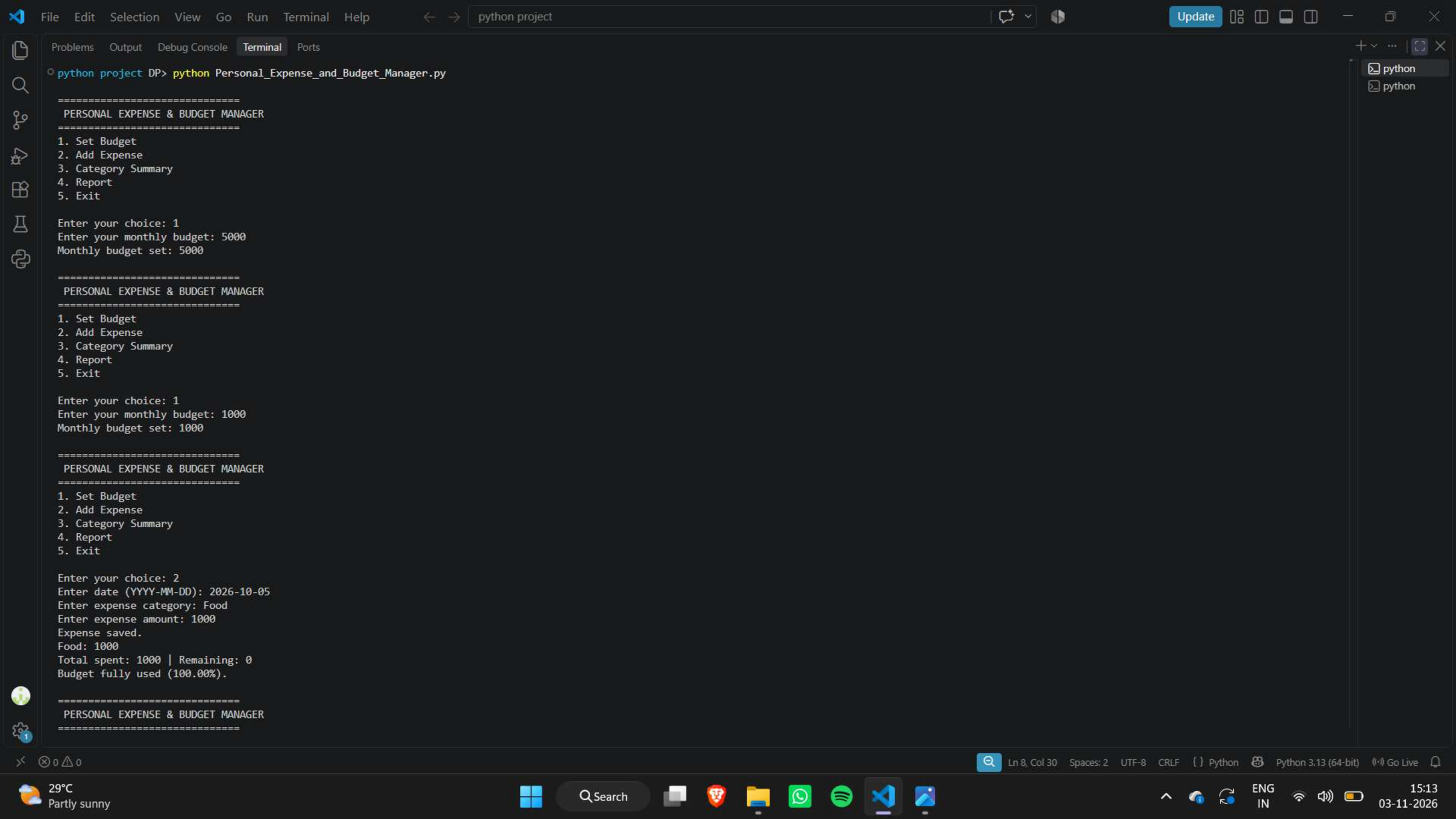
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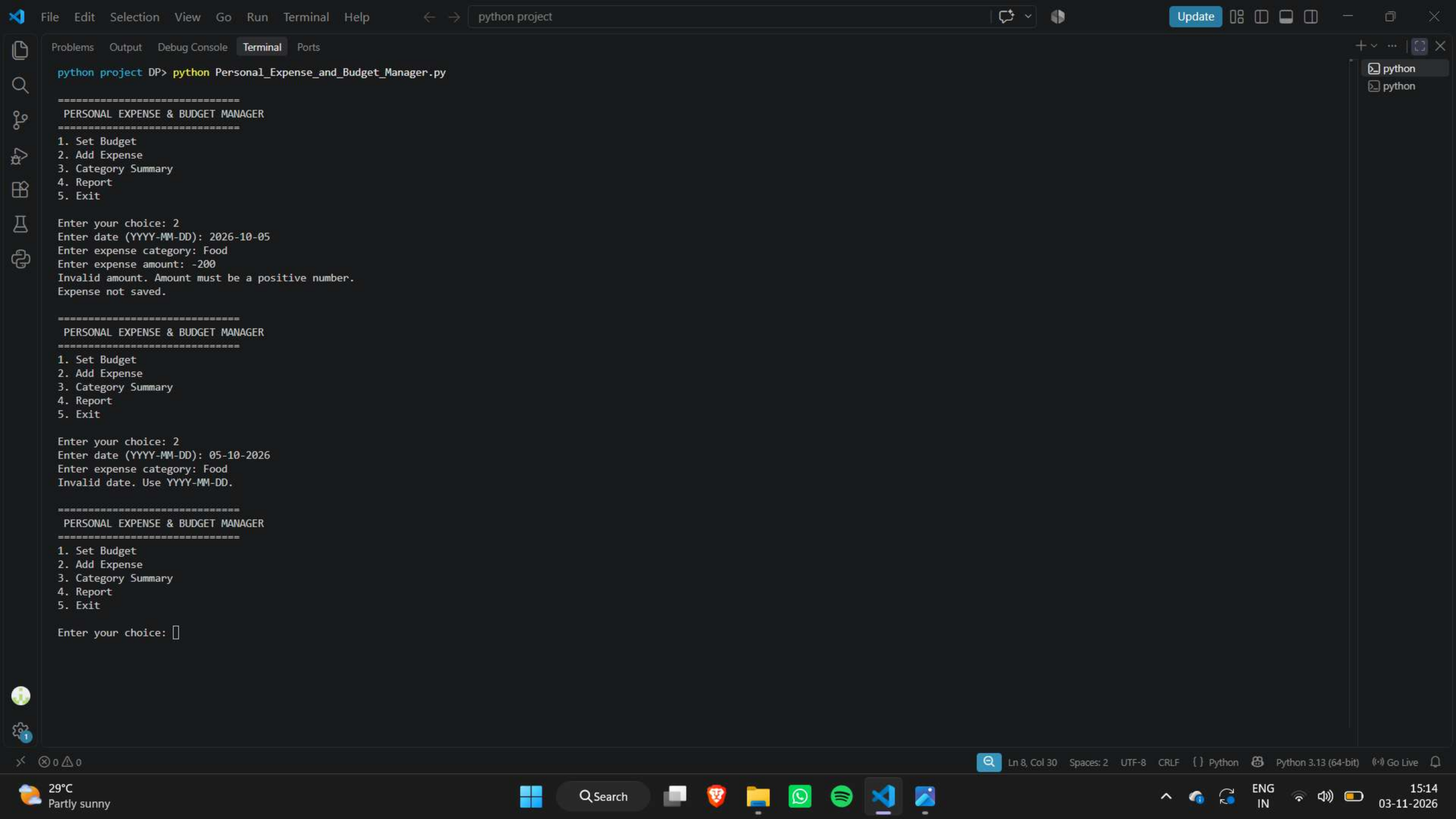
PS No.	42
Project Title	**Personal Expense & Budget Manager**
Problem Statement	Build a tool that stores expenses in a CSV file, lets the user set a monthly budget, and reports category-wise spending, the remaining budget and warnings when the budget is almost used or exceeded. **Language:** Python 3 **Concepts:** Functions, Dictionaries, CSV, Pandas **Note:** Expenses are saved in expenses.csv with columns Date, Category, Amount. Menu: 1 Set budget, 2 Add expense, 3 Category summary, 4 Report. A warning is shown when 80% of the budget is used and 'Budget exceeded' above 100%.
Test Cases / Case Statements (4 cases)	

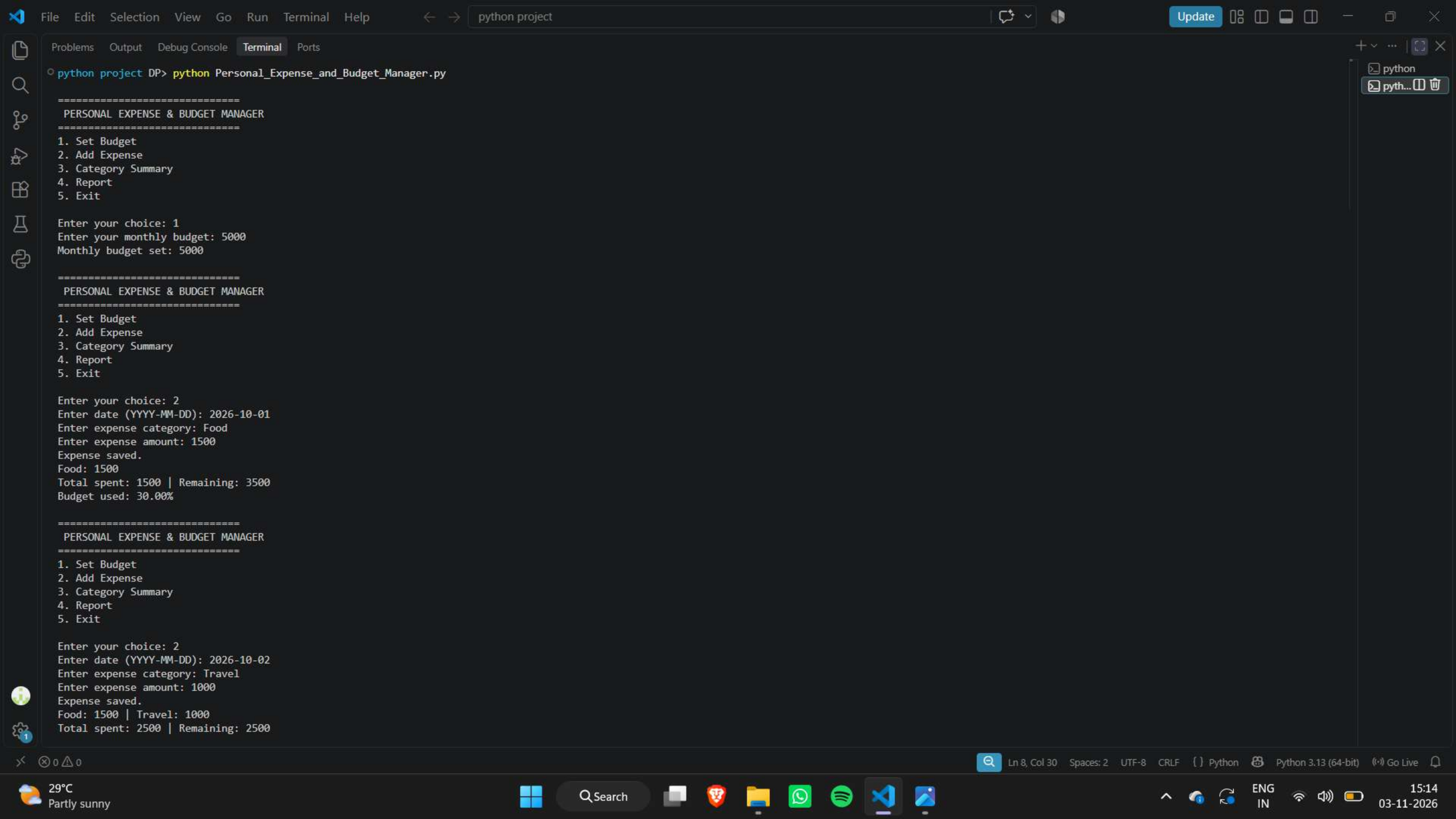
INPUT	EXPECTED OUTPUT
Test Case 1: Normal Input	
Menu option 1 (Set budget) Budget: 5000	Monthly budget set: 5000
Test Case 2: Boundary / Minimum Input	
Budget: 1000 Menu option 2 (Add) 2026-10-05, Food, 1000	Expense saved. Total spent: 1000 | Remaining: 0 Budget fully used (100.00%).
Test Case 3: Invalid Input	
(a) Add: 2026-10-05, Food, -200 (b) Add: 05-10-2026, Food, 200	(a) Invalid amount. Amount must be a positive number. Expense not saved. (b) Invalid date. Use YYYY-MM-DD.
Test Case 4: Different Valid Scenario	
Budget: 5000 Expenses: Food 1500, Travel 1000, Food 1700 Menu option 4 (Report)	Food: 3200 | Travel: 1000 Total spent: 4200 | Remaining: 800 Warning: 84.00% of the budget used.

Bare Minimum Requirements	1. Write the solution in Python 3 (one .py file) and run it from the console. 2. Store and read the expenses using a CSV file and a Pandas DataFrame. 3. Use groupby() to show the category-wise spending. 4. Calculate the total spent, the remaining budget and the percentage of the budget used. 5. Show a warning at 80% and a 'Budget exceeded' message above 100%, and validate the date and the amount.
Instruction	1. Understand the problem statement and define the problem clearly. 2. Develop the solution using proper logic and programming concepts. 3. Implement all required features and pass the given test cases. 4. Test and debug the project to ensure correct output. 5. Prepare the documentation/report and upload the code and report to GitHub. 6. Complete and submit the project within the given deadline.

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python project
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Problems Output Debug Console Terminal Ports
python project DP> python Personal_Expense_and_Budget_Manager.py
==========
PERSONAL EXPENSE & BUDGET MANAGER
==========
1. Set Budget
2. Add Expense
3. Category Summary
4. Report
5. Exit
Enter your choice: 1
Enter your monthly budget: 5000
Monthly budget set: 5000
==========
PERSONAL EXPENSE & BUDGET MANAGER
==========
1. Set Budget
2. Add Expense
3. Category Summary
4. Report
5. Exit
Enter your choice: 1
Enter your monthly budget: 1000
Monthly budget set: 1000
==========
PERSONAL EXPENSE & BUDGET MANAGER
==========
1. Set Budget
2. Add Expense
3. Category Summary
4. Report
5. Exit
Enter your choice: 2
Enter date (YYYY-MM-DD): 2026-10-05
Enter expense category: Food
Enter expense amount: 1000
Expense saved.
Food: 1000
Total spent: 1000 | Remaining: 0
Budget fully used (100.00%).
==========
PERSONAL EXPENSE & BUDGET MANAGER
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python project
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Problems Output Debug Console Terminal Ports
python project DP> python Personal_Expense_and_Budget_Manager.py
==============================
PERSONAL EXPENSE & BUDGET MANAGER
==============================
1. Set Budget
2. Add Expense
3. Category Summary
4. Report
5. Exit
Enter your choice: 2
Enter date (YYYY-MM-DD): 2026-10-05
Enter expense category: Food
Enter expense amount: -200
Invalid amount. Amount must be a positive number.
Expense not saved.
==============================
PERSONAL EXPENSE & BUDGET MANAGER
==============================
1. Set Budget
2. Add Expense
3. Category Summary
4. Report
5. Exit
Enter your choice: 2
Enter date (YYYY-MM-DD): 05-10-2026
Enter expense category: Food
Invalid date. Use YYYY-MM-DD.
==============================
PERSONAL EXPENSE & BUDGET MANAGER
==============================
1. Set Budget
2. Add Expense
3. Category Summary
4. Report
5. Exit
Enter your choice:
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Problems Output Debug Console Terminal Ports
python project DP> python Personal_Expense_and_Budget_Manager.py
==========
PERSONAL EXPENSE & BUDGET MANAGER
==========
1. Set Budget
2. Add Expense
3. Category Summary
4. Report
5. Exit
Enter your choice: 1
Enter your monthly budget: 5000
Monthly budget set: 5000
==========
PERSONAL EXPENSE & BUDGET MANAGER
==========
1. Set Budget
2. Add Expense
3. Category Summary
4. Report
5. Exit
Enter your choice: 2
Enter date (YYYY-MM-DD): 2026-10-01
Enter expense category: Food
Enter expense amount: 1500
Expense saved.
Food: 1500
Total spent: 1500 | Remaining: 3500
Budget used: 30.00%
==========
PERSONAL EXPENSE & BUDGET MANAGER
==========
1. Set Budget
2. Add Expense
3. Category Summary
4. Report
5. Exit
Enter your choice: 2
Enter date (YYYY-MM-DD): 2026-10-02
Enter expense category: Travel
Enter expense amount: 1000
Expense saved.
Food: 1500 | Travel: 1000
Total spent: 2500 | Remaining: 2500
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Problems Output Debug Console Terminal Ports

```
python project DP> python Personal_Expense_and_Budget_Manager.py
============================
1. Set Budget
2. Add Expense
3. Category Summary
4. Report
5. Exit

Enter your choice: 2
Enter date (YYYY-MM-DD): 2026-10-02
Enter expense category: Travel
Enter expense amount: 1000
Expense saved.
Food: 1500 | Travel: 1000
Total spent: 2500 | Remaining: 2500
Budget used: 50.00%

============================
 PERSONAL EXPENSE & BUDGET MANAGER
============================
1. Set Budget
2. Add Expense
3. Category Summary
4. Report
5. Exit

Enter your choice: 2
Enter date (YYYY-MM-DD): 2026-10-03
Enter expense category: Food
Enter expense amount: 1700
Expense saved.
Food: 3200 | Travel: 1000
Total spent: 4200 | Remaining: 800
Warning: 84.00% of the budget used.

============================
 PERSONAL EXPENSE & BUDGET MANAGER
============================
1. Set Budget
2. Add Expense
3. Category Summary
4. Report
5. Exit

Enter your choice: 4
Food: 3200 | Travel: 1000
Total spent: 4200 | Remaining: 800
Warning: 84.00% of the budget used.
```